Why the Claimed BB(7) Champion Halts: the Induction, Written Out on Paper

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12 November 2026
working note; the halting part is now proved in Coq (`BB7Halts.v`)

Abstract

The machine `1RB0RG_1RC1RB_0RD1RZ_0LE0LD_0LF1LE_1RA1LF_1LC0RB` (found by Jeff Ketchersid) is listed as the BB(7) champion, with the claim marked as not independently verified. In earlier notes we proved in Coq that the machine reaches a clean tape with 19 units, that one *move* rule describes each macro step for every gap width, and that a *gate* rule halts the machine. What was missing is a proof that the run uses only these rules until the end. This note gives that proof on paper. It is an induction on the gap width. It also gives the exact value of the last block as a ladder of functions $G_1, G_2, \ldots$ The argument is checked by computer on every case small enough to compute. The halting part has since been proved in Coq (`BB7Halts.v`, no axioms): the machine halts from a blank tape. The exact value of the last block is still a paper result.

1 What is already proved

A *clean tape* is written

$$\big(P \mid (g_1, Z_1), (g_2, Z_2), \ldots, (g_r, Z_r)\big).$$

It means: a block of P ones, then g_1 zeros, then a block of Z_1 ones, then g_2 zeros, and so on, with the head in state G at the left of the first block. We call P the *working block*, each Z_i a *counter*, and each g_i a *gap*. A gap of $2j$ cells is said to have *width* j.

Three facts are proved in `BB7Moves.v`.

Start (`start`). From a blank tape, after 3,954 steps the tape is $\big(16 \mid (4,2), (2,2)^{18}, (2,1)\big)$. Here $(2,2)^{18}$ means eighteen copies. So there are 19 counters equal to 2 (the *units*) and a last counter 1.

Move (`move`). Let the first gap have width $j \geq 1$ and let $m = 3P + 3 - 2j$. If $m \geq 1$ and $Z \geq 2$, then

$$\underbrace{1^P\, 0^{2j}\, 1^Z\, R}_{\text{before}} \quad \longrightarrow \quad \underbrace{(1100)^{j-1}\, 1^N\, 00\, 1^{Z-2}\, R}_{\text{after}}, \qquad N = 3P + 8 - 2j.$$

R is any rest of the tape. We say the move is *in range* when $m \geq 1$.

Gate (`halts_gen`). If the tape is $\big(P \mid (2j, 1)\big)$ with nothing after the last 1, and $m = 3P+3-2j$ is odd and at least 1, the machine halts.

2 The move, read on clean tapes

Reading the move rule in the clean-tape notation gives four cases. Let T be any list of (gap, counter) pairs and $N = 3P + 8 - 2j$.

Case	Before	After
T (triple), $j = 1$, $Z \geq 4$	$\big(P \mid (2, Z), T\big)$	$\big(3P + 6 \mid (2, Z - 2), T\big)$
T′ (counter ends), $j = 1$, $Z = 2$	$\big(P \mid (2, 2), (g, X), T\big)$	$\big(3P + 6 \mid (g + 2, X), T\big)$
S (swap), $j \geq 2$, $Z \geq 4$	$\big(P \mid (2j, Z), T\big)$	$\big(2 \mid (2, 2)^{j-2}, (2, N), (2, Z - 2), T\big)$
S′ (counter ends), $j \geq 2$, $Z = 2$	$\big(P \mid (2j, 2), (g, X), T\big)$	$\big(2 \mid (2, 2)^{j-2}, (2, N), (g + 2, X), T\big)$

In words. With a single gap the working block triples. With a wider gap the tripled block becomes a new counter N, a fresh working block of 2 is planted, and $j - 2$ extra counters of 2 are planted between them. When a counter is used up, its two gaps join into one wider gap.

Lemma 1 (Even numbers)**.** *If P is even and every counter except possibly the last is even, the same is true after any move.*

Proof. $3P + 6$ is even. $N = 3P + 8 - 2j$ is even. The planted values are 2. $Z - 2$ is even. □

After the start, $P = 16$ and the counters are $2, \ldots, 2, 1$. So from now on *P is always even, every counter but the last is even, and the last counter stays* 1 *and stays last.*

3 The ladder functions

Definition 1. For even P define

$$G_1(P) = 3P + 6, \qquad G_j(P) = G_{j-1}^{(3P+8-2j)/2}\big(p_{j-1}\big) \quad (j \geq 2),$$

where $f^{\,n}$ means "apply f n times", and the constants p_k are

$$p_1 = 2, \qquad p_k = G_{k-1}(p_{k-1}) \quad (k \geq 2).$$

So $p_2 = 12$, $p_3 = G_2(12) = G_1^{20}(2) = 5 \cdot 3^{20} - 3$, and for example $G_2(P) = G_1^{(3P+4)/2}(2) = 5 \cdot 3^{(3P+4)/2} - 3$. G_1 is linear, G_2 is an exponential, G_3 is a tower of exponentials, and each further G_j is one level higher.

4 The consume lemma

Lemma 2 (Consume)**.** *Let $1 \leq j \leq 199$. Let P and Z be even, $Z \geq 2$, and let the first move be in range ($3P + 3 - 2j \geq 1$). Then for every next pair (g, X) and every T,*

$$\big(P \mid (2j, Z),\, (g, X),\, T\big) \implies^{*} \big(G_j^{Z/2}(P) \mid (g + 2j,\, X),\, T\big),$$

and every move used on the way is in range.

In words: a counter Z behind a gap of width j is used up completely; the working block becomes G_j applied $Z/2$ times; and the next gap grows by $2j$.

Proof. By induction on j.

$j = 1$. Apply case T $(Z - 2)/2$ times, then case T′ once. Each application replaces P by $3P + 6 = G_1(P)$, and the last one joins the gaps to $g + 2$. Every move has gap width 1 and $P \geq 2$, so $m = 3P + 1 \geq 7$.

$j \geq 2$. We show one *round*:

$$\big(P \mid (2j, Z), \dots\big) \implies^* \begin{cases} \big(G_j(P) \mid (2j, Z-2), \dots\big) & Z \geq 4, \\ \big(G_j(P) \mid (g+2j, X), T\big) & Z = 2. \end{cases} \tag{1}$$

Repeating the round $Z/2$ times gives the lemma.

The first move is case S or S′. It gives

$$\big(2 \mid (2,2)^{j-2},\ (2, N),\ E, \dots\big), \qquad N = 3P + 8 - 2j,$$

where $E = (2, Z-2)$ if $Z \geq 4$ and $E = (g+2, X)$ if $Z = 2$.

The planted counters. The working block is $p_1 = 2$ and it meets a counter 2 behind a gap of width 1. By the lemma for width 1 the block becomes $G_1(p_1) = p_2$ and the next gap becomes $2 + 2 = 4$. Now p_2 meets a counter 2 behind a gap of width 2; by the lemma for width 2 the block becomes p_3 and the next gap becomes 6. Going on like this, the k-th planted counter is met by p_k behind a gap of width k, for $k = 1, \dots, j-2$, and after all of them the tape is

$$\big(p_{j-1} \mid (2(j-1), N),\ E, \dots\big).$$

The big counter. By the lemma for width $j-1$ applied to the counter N, the block becomes $G_{j-1}^{N/2}(p_{j-1})$, which is $G_j(P)$ by Definition 1, and the gap in front of E grows by $2(j-1)$. If $E = (2, Z-2)$ the new gap is $2j$. If $E = (g+2, X)$ the new gap is $g + 2j$. This is (1).

Range. We must check $3P' + 3 - 2j' \geq 1$ for each working block P' that meets a gap of width j':

- $p_1 = 2$ meets width 1: $m = 7$.
- $p_2 = 12$ meets width 2: $m = 35$.
- p_k with $k \geq 3$ meets width k. By Lemma 3 below $p_k \geq 132$, so $m \geq 399 - 2k \geq 1$ because $k \leq 199$.
- After a round, $G_j(P)$ meets width j again. By Lemma 3, $G_j(P) \geq 132$, so again $m \geq 1$.

Also $N = m + 5 \geq 6$, so N is an even counter with $N/2 \geq 3$. □

Lemma 3 (Blocks are big enough)**.** *For $j \geq 2$, whenever the first move is in range, $G_j(P) \geq 132$. Hence $p_k \geq 132$ for $k \geq 3$.*

Proof. $G_j(P) = G_{j-1}^n(p_{j-1})$ with $n = N/2 \geq 3$. For $j = 2$ this is $G_1^n(2) \geq G_1^3(2) = 132$, since G_1 is increasing and $G_1(x) > x$. For $j \geq 3$ the last of the n applications is a value of G_{j-1} on an in-range block, which is at least 132 by induction. □

Remark 1. Lemmas 2 and 3 refer to each other. They are proved together by one induction on j: the proof for width j uses both lemmas only for smaller widths, and Lemma 3 for width j itself only after the first round of width j is done.

Remark 2. The bound $j \leq 199$ is only there because Lemma 3 uses the crude number 132. The blocks are in fact far larger. The real run needs widths up to 21.

5 The theorem

Theorem 1. *Let $1 \le u \le 197$ and let $P_0 \ge 2$ be even. Define*

$$Q_0 = P_0, \qquad Q_k = G_{k+1}(Q_{k-1}) \quad (k = 1, \dots, u).$$

Then the machine, started on the clean tape $\big(P_0 \mid (4,2),\,(2,2)^{u-1},\,(2,1)\big)$, uses only in-range moves, reaches

$$\big(Q_u \mid (2(u+2),\,1)\big),$$

and halts there.

Proof. The first unit is behind a gap of width 2. By Lemma 2 with $j = 2$ and $Z = 2$ it is used up, the block becomes $G_2(Q_0) = Q_1$, and the gap of the second unit grows from 2 to 6, width 3. By the lemma with $j = 3$ the block becomes Q_2 and the third unit's gap becomes 8. In general the k-th unit is met by Q_{k-1} behind a gap of width $k+1$, and is used up by the lemma with $j = k+1$. The first moves are in range: $Q_0 \ge 2$ meets width 2 ($m = 3P_0 - 1 \ge 5$), and $Q_k \ge 132$ for $k \ge 1$.

After the last unit the tape is $\big(Q_u \mid (2(u+2), 1)\big)$. Q_u is even by Lemma 1, so $m = 3Q_u + 3 - 2(u+2)$ is odd, and it is at least 1 because $Q_u \ge 132$. The gate rule applies and the machine halts. □

Corollary 1 (The champion)**.** *The machine halts from a blank tape. The widest gap in a move is of width* 20*, the halt is at width* 21*, and the last working block is*

$$Q_{19} = G_{20}\big(G_{19}(\cdots G_3(G_2(16))\cdots)\big).$$

Proof. The start rule gives the tape of Theorem 1 with $P_0 = 16$ and $u = 19$. □

The first value is $Q_1 = G_2(16) = 5 \cdot 3^{26} - 3 = 12{,}709{,}329{,}141{,}642$. The second, $Q_2 = G_3(Q_1)$, is G_2 applied 19,063,993,712,464 times to 12: a tower of that many exponentials. We do not work out a closed bound here. The claimed bound of the form $3 \uparrow^{20} 3$ has the same shape as a ladder with twenty levels, but comparing the two is left for later.

6 Computer checks

All of these are in the repository, in `bb7/`.

Check	Result
Move and halt words against the machine, digit for digit, widths 1 to 21 (`check_wide_gaps.py`)	63 of 63 and 63 of 63 match
Every shape reachable with 1 to 12 units (`check_closure.py`)	no shape without a rule; widest move $u+1$; one halt, at width $u+2$
Lemma 2 against the exact rules, ten small cases (`check_induction.py`)	10 of 10 agree (values up to 23,478 digits)
Theorem 1 with $u = 1$ against the real machine	$P_0 = 2$: halts at step 18,310,165 as predicted, last block $1212 = G_2(2)$. $P_0 = 4$: halts at step 13,318,711,129 as predicted, last block $32802 = G_2(4)$
Q_1 and the counter $3Q_1 + 2$ against the macro run	both equal

No case with two or more units can be run to the end; the numbers are too large.

7 What is proved and what is not

- **Proved in Coq** (`BB7Moves.v`, `BB7Halts.v`, Coq 8.18, no axioms): the start, the move for every width, the gate, the consume lemma for widths 1 to 199 (as "the counter is used up, the next gap grows by $2j$, and the new block is even and large enough"), and

  `champion_halts`: there is an n such that the machine, started on a blank tape, is in the halt state after n steps.

  Also `champion_gate`: the run reaches one block, a gap of 42 cells (width 21) and the last 1.

- **Proved on paper only**: the exact values, that is, that the new block after a consume is $G_j^{Z/2}(P)$ and that the last block is $G_{20}(\cdots G_2(16))$. The Coq proof shows the block exists, is even and is at least 132; it does not name it.

- The Coq version of Theorem 1 (`units_halt`) asks for $P_0 \geq 12$ instead of $P_0 \geq 2$. The champion has $P_0 = 16$.

- **Not done**: the halting time, the number of ones left, and a comparison with the claimed bound.

- We do not know whether this argument is already known to the discoverer or to others.

8 How the Coq proof is arranged

1. Numbers are written by their halves: the working block is $2q$ ones and a counter is $2h+2$ ones. Then "even" needs no side condition.

2. `consume1`: width 1, by induction on h.

3. `planted`: a row of t units behind a gap of width $a+1$ is used up at widths $a+1, \ldots, a+t$, given the consume lemma for those widths.

4. `round`: one round at width $j \geq 2$: the move, then `planted` for the $j-2$ planted units, then the consume lemma at width $j-1$ for the big counter.

5. `consume_all`: all widths up to 199, by induction on the width with an inner induction on h.

6. `units_to_gate`: `planted` once more, now for the 19 real units ($a = 1$). Then `halts_gen` gives `champion_halts`.

Every lemma carries a free rest-of-tape, as the move theorem does. That is what lets the consume lemma be used inside itself.